Iterative Semi-Supervised Learning With Few-Shot Samples for Coastal Wetland Land Cover Classification

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***Abstract*— A novel approach is proposed in this study that combines superpixel (SP) segmentation and multiclassifier ensemble learning (EL) to address the limited availability of labeled samples in coastal wetland land cover classification. First, the SP segmentation technique is employed to partition unknown samples into multiple homogeneous regions, thereby facilitating the effective capture of spatial information pertaining to land cover. Subsequently, a multiclassifier EL strategy is employed within these regions to process the samples, effectively leading to a reduction in classification errors and an improvement in accuracy. To enhance the performance of semi-supervised learning (SSL), a sample iteration selection metric is introduced to optimize the training samples based on the consistency of sample types within homogeneous regions and the results obtained from the multiclassifier ensemble, thus enhancing the reliability of pseudo-labels. Additionally, multiscale SP segmentation is utilized to augment the ensemble strategy for samples in order to reduce the necessity for hyperparameter adjustments and increase the automation and reliability of the model. Overall, the accuracy of coastal wetland classification is improved by this approach while simultaneously mitigating the complexity of SSL in terms of hyperparameter tuning. The effectiveness of the proposed approach has been assessed through experiments conducted on three GF-5 hyperspectral images of coastal wetlands in China. In particular, the proposed methods provide superior performance compared with the state-of-the-art classification methods.**

***Index Terms*— Coastal wetland, ensemble learning (EL), hyperspectral, land cover classification, limited samples, superpixel (SP) segmentation.**

I. Introduction

COASTAL wetland land cover classification is typically encompassed by a variety of ecosystems and land types, which are considered crucial for ecological balance and biodiversity [1], [2]. Therefore, accurate categorization of coastal wetland land cover types is seen as beneficial for more effective protection and management of these ecosystems, promotion of ecological balance, biodiversity, and facilitation of sustainable development for human societies [3], [4], [5]. The utilization of hyperspectral image (HSI) for fine classification of wetlands is known to offer advantages, as it can provide richer spectral information, aiding in the accurate identification of different wetland types and the monitoring of ecological changes [6]. However, when HSIs are employed for coastal wetland land cover classification, a significant challenge arises i.e., unavailability of a substantial amount of ground-truth sample data to train and validate classification models. Because coastal wetlands comprise diverse ecosystems and land types, the issue of limited labeled samples can greatly reduce the accuracy and generalization capability of classification models [7], [8]. The content mentioned is the "few-shot" problem in HSI classification. Few-shot learning (FSL) is a field in machine learning that aims to construct highly accurate classification or regression models using very few training data, which is very useful in overcoming the limited sample number problem in hyperspectral remote sensing image classification. FSL methods can be mainly divided into three categories: hand-crafted rules (HCR) methods, ensemble learning-based (EL) methods, and co-training methods. HCR methods enhance data through geometric transformations of images, but their applicability is limited due to their dependence on the invariances of specific images. EL methods increase the diversity of training data by resampling them to different spaces or feature spaces, but they may not be suitable for hyperspectral data as they may lose band information. Co-training methods utilize high-confidence samples from unlabeled data for expansion by adding them to the training data for further training. Overall, these methods each have their strengths and limitations, and therefore, in the task of few-shot hyperspectral remote sensing image classification, combining multiple methods, especially EL methods, is a promising strategy to improve classification accuracy. In related research, the proposed refined prototypical contrastive learning (RPCL)-FSL method enhances the stability of prototypes and mitigates domain shift, improving few-shot HSI classification performance [9]. The deep contrastive learning network (DCLN) method, which integrates deep con-

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trastive learning with pseudo-labeling and iterative training, surpasses existing state-of-the-art techniques in HSI classification when labeled data is scarce [10]. The class-aligned and class-balancing generative domain adaptation (CCGDA) method integrates class alignment and class balancing to improve domain adaptation for HSI classification, addressing the limitations of existing domain adaptive techniques [11]. The meta-reinforcement (MR)-Selection method applies zero-shot band selection to HSIs by leveraging metaknowledge through a shared agent, resulting in a more efficient and generalizable approach to band selection [12]. However, the above methods have some limitations when dealing with small sample HSI classification problems: the HCR method relies on the invariance of specific images, resulting in low scalability and difficulty in adapting to most classification problems, and it is challenging to extract effective domain information when processing HSIs; although the EL-based methods increase the diversity of training data through resampling, they perform poorly when the initial number of training samples is limited and may lose band information in HIS or lead to the loss of categories when using bootstrap methods; the co-training-based methods, while leveraging the potential of unlabeled data, are too dependent on the accuracy of the initial classification results. In small sample scenarios, the low precision of the initial training samples can lead to a large number of incorrect classification results, thereby generating noise samples and reducing classification accuracy.

Another effective method for addressing the small sample problem in hyperspectral wetland classification is the application of semi-supervised learning (SSL) [13]. SSL is a machine learning technique designed to tackle the challenge of limited labeled data. In traditional SSL, models are trained and predictions are made using exclusively labeled data [14], [15], [16]. However, in real-world scenarios, obtaining a large quantity of labeled data is constrained by factors such as cost and time. The fundamental concept underlying SSL is to maximize the utility of unlabeled data to enhance model performance. SSL can be broadly categorized into three principal classes [17].

A. Generative Model-Based Approaches [18]

These approaches employ generative models such as generative adversarial networks (GANs) [19] or variational autoencoders (VAEs) [20], [21] to generate synthetic samples from unlabeled data. Generative models [22], [23], [24] are trained to capture the data distribution and to produce samples that closely resemble real data. Subsequently, these synthetic samples can be integrated with labeled data to train a supervised learning model. The effectiveness of generative models directly impacts the quality of the synthetic samples, thereby exerting an influence on the outcomes of SSL [25]. However, methods reliant on generative models for SSL may necessitate a substantial quantity of unlabeled data and computational resources and may introduce synthetic samples of lower quality.

B. Graph-Based Approaches [26]

These methods are tailored for data with a graph-based structure, such as graphs representing user-item relationships in social networks or recommendation systems [27]. They leverage the topological structure of the graph and relationships between nodes to propagate labels or generate node embeddings. Commonly used graph-based SSL techniques encompass label propagation (LP), graph convolutional networks (GCNs), and graph generative models, among others. These methods typically exhibit prowess in harnessing information pertaining to interrelationships among data points. Nevertheless, graph-based SSL [28] methods presume a strong underlying graph structure in the data, which may result in suboptimal performance on non-graph-structured data and require more computational resources.

C. Self-Training Approaches [29]

Self-training methods involve initially training a supervised learning model on labeled data, followed by utilizing this model to make predictions on unlabeled data [30]. Predictions with high confidence are subsequently incorporated into the labeled dataset, and model training is continued. This iterative process can be repeated multiple times. Self-training methods are often characterized by their simplicity, but they may exhibit vulnerability when confronted with noisy data and high label uncertainty [31]. Furthermore, self-training methods are susceptible to noise in the data and pseudo-labeling, necessitating judicious hyperparameter tuning to mitigate overfitting.

Based on the above analysis, it can be seen that SSL methods primarily face three key scientific challenges. First, there is the issue of how to effectively select unlabeled data for sample augmentation, which involves selecting high-confidence samples and addressing noise data problems. Second, there is the challenge of determining the hyperparameters for various SSL models. The last challenge is how to increase the confidence of augmented samples and how to reuse all unlabeled samples. To solve the above problems, a novel wetland HSI classification method with superpixel (SP) segmentation [32], [33], [34] and multiclassifier EL is proposed in this article. This method aims to effectively overcome the challenges in traditional approaches. The homogeneity of samples of the same type in the spatial domain is fully utilized, and a series of steps are taken to classify unknown samples. First, in order to make full use of the homogeneity feature, SP segmentation is employed to partition unknown samples into multiple homogenous regions. This can cluster similar pixels to better extract the spatial distribution information of land cover. Subsequently, within each homogenous region, the approach employs the multiclassifier EL method to classify unknown samples. Multiclassifier EL integrates decisions from multiple classifiers to reduce classification errors. To ensure the reliability of pseudo-labels, an indicator for sample iterative selection is introduced, which considers the consistency of sample types within homogenous regions and the results of multiclassifier EL. This helps further improve the quality of pseudo-labels. Furthermore, to address the common issue of hyperparameter tuning in SSL, the article introduces the concepts of dictionary learning and EL to achieve consistency learning and ensemble on the results of multiscale SP segmentation-augmented samples. This step mitigates the need for hyperparameter selection and tuning, making the method more automated and reliable.

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The specific contributions of the article can be summarized as follows.

1) *Homogeneous Sample Segmentation and Spatial Information Capture:* SP segmentation techniques are employed in this method to partition unknown samples into multiple homogeneous regions. This step helps with the aggregation of neighboring similar pixels, thereby allowing the spatial distribution information of land cover to be effectively captured. This means that during the classification process, the spatial relationships among samples in the geographic space are considered, not just pixel-level information, leading to improved classification accuracy.
2) *Advantages of Multiclassifier EL:* Multiclassifier EL is utilized to handle unknown samples within homogeneous regions. It does not rely on a single classifier but integrates decisions from multiple classifiers. Such an approach can result in the reduction of classification errors and the enhancement of classification robustness. For instance, if one classifier performs poorly in a specific region, its shortcomings can be compensated for by other classifiers, thereby improving overall classification performance.
3) *Enhancement of Pseudo-Label Quality:* An indicator for sample iterative selection is introduced to continuously improve the quality of pseudo-labels. This indicator takes into account the consistency of sample types within homogeneous regions and the results of multiclassifier EL. Through this approach, pseudo-labels become more reliable. This method effectively addresses the challenges of unlabeled samples, particularly in the presence of noisy data.

This article is organized as follows. In Section II, the related works are introduced. Section III presents the proposed homogeneous region-constrained multiclassifier consensus sample selection strategy (HRC-MS3) and iterative multiscale sample expansion with weighted dictionary learning (IMSE-WDL). The results of the experiment are demonstrated in Section IV. Section V discusses parameter sensitivity, effectiveness, and other experiments of the proposed algorithms in detail. Finally, the conclusions of this study are drawn in Section VI.

II. Related Works

A. Entropy Rate Superpixel Segmentation (ERS)

The ERS algorithm is employed in this article for the segmentation of hyperspectral remote sensing images, resulting in regions with homogeneity. The algorithm is described as follows.

The ERS constructs a graph $\boldsymbol{G} = (\boldsymbol{V}, \boldsymbol{E})$, where pixels represent vertices and edge weights are determined by a similarity matrix $\boldsymbol{W}$, indicating pairwise similarities. The weight of the edge $e_{i,j}$ between vertexes v_i and v_j is defined using the Gaussian kernel as

$$w_{i,j} = \exp\left(\frac{d(v_i, v_j)^2}{2\sigma^2}\right) \tag{1}$$

where $d(v_i, v_j)$ is defined as the intensity difference multiplied by the spatial distance and σ is the kernel parameter.

The objective is to choose a subset of edges, denoted as $\boldsymbol{A} \in \boldsymbol{E}$, such that the resulting graph $\boldsymbol{G} = (\boldsymbol{V}, \boldsymbol{A})$ consists of precisely $\boldsymbol{K}$ connected subgraphs. The objective function is given as

$$\max_{\boldsymbol{A}} H(\boldsymbol{A}) + \lambda B(\boldsymbol{A}) \tag{2}$$

where λ is a regularization parameter used to balance weights.

Where $H(\boldsymbol{A})$ is the entropy rate of random walks for graph $\boldsymbol{G}$ represented by

$$H(\boldsymbol{A}) = \sum_i u_i \sum_i p_{i,j}(\boldsymbol{A}) \log\left(p_{i,j}(\boldsymbol{A})\right) \tag{3}$$

$$p_{i,j}(\boldsymbol{A}) = \begin{cases} \dfrac{w_{i,j}}{w_i}, & i \neq j,\ e_{i,j} \in \boldsymbol{A} \\ 0, & i \neq j,\ e_{i,j} \notin \boldsymbol{A} \\ 1 - \dfrac{\sum_{k:e_{i,j} \in \boldsymbol{A}} w_{i,j}}{w_i}, & i = j \end{cases} \tag{4}$$

$$w_i = \sum_{k:e_{i,k} \in \boldsymbol{E}} w_{i,k} \tag{5}$$

where $B(\boldsymbol{A})$ is the balancing function, defined as

$$B(\boldsymbol{A}) = -\sum_i pz_{\boldsymbol{A}}(i) \log\left(pz_{\boldsymbol{A}}(i)\right) - N_{\boldsymbol{A}} \tag{6}$$

$$pz_{\boldsymbol{A}}(i) = \frac{|\boldsymbol{S}_i|}{|\boldsymbol{V}|}, \quad i = \{1, \ldots, N_{\boldsymbol{A}}\} \tag{7}$$

where $\boldsymbol{S}_i$ is the balanced cluster.

Finally, an iterative greedy algorithm is employed to address the graph optimization problem outlined in (2).

B. Base Classifiers Used in the Article

Collaborative representation-based classification (CRC), originally developed for face recognition, has gained popularity in HSI classification for its outstanding performance. Consider a training sample denoted by $\boldsymbol{X} = [x_1, \ldots, x_n]$, which consists of n features. The training set is a matrix $\boldsymbol{X}$ of size $n \times N$. The M testing samples, denoted as $\boldsymbol{Y} = [y_1, \ldots, y_n]$, are assigned with class labels from the set $\{1, \ldots, C\}$, where C represents the total number of classes. The CRC coefficient $\boldsymbol{\alpha}$ is obtained

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha}^*} \left\|\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\alpha}^*\right\|_2^2 + \lambda \left\|\boldsymbol{\alpha}^*\right\|_2^2 \tag{8}$$

where $\|\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\alpha}^*\|_2^2$ is the reconstruction error, λ is the regularization parameter, and $\lambda\|\boldsymbol{\alpha}^*\|_2^2$ controls the complexity of $\boldsymbol{\alpha}^*$. By zeroing out the partial derivative, the analytical solution can be derived

$$\boldsymbol{\alpha} = \left(\boldsymbol{X}^T\boldsymbol{X} + \lambda \boldsymbol{I}\right)^{-1}\boldsymbol{X}^T\boldsymbol{Y} \tag{9}$$

where $\boldsymbol{I}$ denotes an identity matrix. To tackle the challenge posed by the linear inseparability of pixels in image classification, researchers have introduced an approach known as kernel collaborative classification (KCRC) [35]. This method reformulates the optimization problem as follows:

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha}^*} \left\|\boldsymbol{\Phi}(\boldsymbol{Y}) - \boldsymbol{\Phi}(\boldsymbol{X})\boldsymbol{\alpha}^*\right\|_2^2 + \lambda \left\|\boldsymbol{\alpha}^*\right\|_2^2 \tag{10}$$

where $\boldsymbol{\Phi}(\boldsymbol{Y})$ and $\boldsymbol{\Phi}(\boldsymbol{X})$ represent the testing samples.

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In addition to introducing the kernel method, Cai et al. also presented a probabilistic CRC (ProCRC), which offers a collaborative representation (CR) approach with a probabilistic interpretation. The coefficient $\boldsymbol{\alpha}$ is obtained

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha}^*} \left\| \boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\alpha}^* \right\|_2^2 + \lambda \left\| \boldsymbol{\alpha}^* \right\|_2^2 + \frac{\eta}{C} \sum_{c=1}^{C} \left\| \boldsymbol{X}\boldsymbol{\alpha}^* - \boldsymbol{X}_c \boldsymbol{\alpha}_c^* \right\|_2^2. \tag{11}$$

The information used for classification usually is the residual corresponding to the cth class

$$r_c(y) = \| \boldsymbol{y} - \boldsymbol{X}_c \boldsymbol{\alpha}_c \|_2^2 \tag{12}$$

$$\boldsymbol{P} = \arg\min_{c} r_c(y) \tag{13}$$

where $r_c(y)$ is the reconstruction residual and $\boldsymbol{P}$ is the final class label. These three heterogeneous CR-based classifiers are selected as the base classifiers for SSL sample ensemble selection in the article.

III. Proposed Method

The use of SP segmentation is motivated by the need to aggregate similar pixels into homogeneous regions based on features such as color, brightness, and texture. This segmentation facilitates the capture of spatial distribution patterns of land cover by creating regions with shared characteristics. By enhancing the meaningfulness of features, SP segmentation improves the accuracy of subsequent classification tasks. The semantic information within these regions, which are larger and more meaningful than individual pixels, aids in the representation of land features or targets. In conclusion, SP segmentation provides a robust starting point for image analysis and offers richer features for subsequent tasks, making it particularly beneficial for applications like object classification and land cover analysis.

A. Homogeneous Region-Constrained Multiclassifier Consensus Sample Selection Strategy (HRC-MS3)

By using CRC, KCRC, and ProCRC, these three different types of CR-based models, the diversity of the models can be increased. This diversity helps to capture different features and patterns in the data, thereby improving the overall classification performance. At the same time, by using heterogeneous CR-based models to generate pseudo labels and incorporating them into the training data, the number of training samples can be effectively increased. This is particularly useful for addressing the issue of scarce labeled data, as it allows the model to learn from more data, thereby improving its generalization ability.

The flowchart of the proposed multiclassifier sample selection method based on ERS is presented in Fig. 1. To achieve homogenous regions, subgraphs $\boldsymbol{Y}_l, i = 1, \ldots, L$ are extracted from testing samples based on the ERS algorithm.

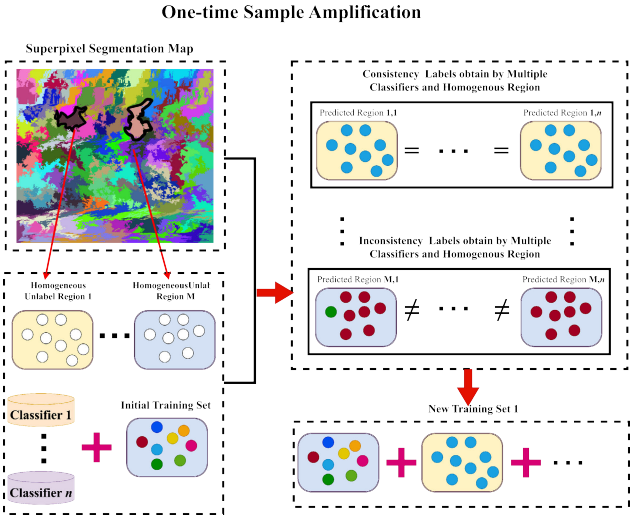


Fig. 1. Flowchart for the proposed HRC-MS3.

The classification results from three different classifiers can be obtained as follows:

$$\begin{cases} \arg\min_{\boldsymbol{\alpha}^*} \left\| \boldsymbol{Y}_l - \boldsymbol{X}\boldsymbol{\alpha}^* \right\|_2^2 + \lambda \left\| \boldsymbol{\alpha}^* \right\|_2^2 \\ \arg\min_{\boldsymbol{\alpha}^*} \left\| \boldsymbol{\Phi}(\boldsymbol{Y}_l) - \boldsymbol{\Phi}(\boldsymbol{X})\boldsymbol{\alpha}^* \right\|_2^2 + \lambda \left\| \boldsymbol{\alpha}^* \right\|_2^2 \\ \arg\min_{\boldsymbol{\alpha}^*} \left\| \boldsymbol{Y}_l - \boldsymbol{X}\boldsymbol{\alpha}^* \right\|_2^2 + \lambda \left\| \boldsymbol{\alpha}^* \right\|_2^2 \\ \qquad + \frac{\eta}{C} \sum_{c=1}^{C} \left\| \boldsymbol{X}\boldsymbol{\alpha}^* - \boldsymbol{X}_c \boldsymbol{\alpha}_c^* \right\|_2^2. \end{cases} \tag{14}$$

Based on (11), the classification results in each subgraph $\boldsymbol{Y}_l$ are obtained $\{\boldsymbol{P}_{clf1,i}, \boldsymbol{P}_{clf2,i}, \boldsymbol{P}_{clf3,i}\}$. For each subgraph $\boldsymbol{Y}_l$, a sample selection index is defined as

$$\boldsymbol{X}_l = \boldsymbol{X} \cup \boldsymbol{P}_{1,i}, \quad \text{if } \boldsymbol{P}_{1,i} = \boldsymbol{P}_{2,i} = \boldsymbol{P}_{3,i}. \tag{15}$$

Then, the new training sample is constructed as

$$\boldsymbol{X}' = \boldsymbol{X}_1 \cup \boldsymbol{X}_2 \cup \cdots \cup \boldsymbol{X}_L. \tag{16}$$

The loss function can be updated as

$$\begin{cases} \arg\min_{\boldsymbol{\alpha}^*} \left\| \boldsymbol{Y}_l - \boldsymbol{X}'\boldsymbol{\alpha}^* \right\|_2^2 + \lambda \left\| \boldsymbol{\alpha}^* \right\|_2^2 \\ \arg\min_{\boldsymbol{\alpha}^*} \left\| \boldsymbol{\Phi}(\boldsymbol{Y}_l) - \boldsymbol{\Phi}(\boldsymbol{X}')\boldsymbol{\alpha}^* \right\|_2^2 + \lambda \left\| \boldsymbol{\alpha}^* \right\|_2^2 \\ \arg\min_{\boldsymbol{\alpha}^*} \left\| \boldsymbol{Y}_L - \boldsymbol{X}'\boldsymbol{\alpha}^* \right\|_2^2 + \lambda \left\| \boldsymbol{\alpha}^* \right\|_2^2 \\ \qquad + \frac{\eta}{C} \sum_{c=1}^{C} \left\| \boldsymbol{X}'\boldsymbol{\alpha}^* - \boldsymbol{X}'_c \boldsymbol{\alpha}_c^* \right\|_2^2. \end{cases} \tag{17}$$

According to (9)–(13), the final classifier result is the majority voting using the results of the new training sample

$$\boldsymbol{P}_E = \text{argmax}\left(\boldsymbol{P}'_{1,i}, \boldsymbol{P}'_{2,i}, \boldsymbol{P}'_{3,i}\right). \tag{18}$$

B. Iterative Multiscale Sample Expansion With Weighted Dictionary Learning (IMSE-WDL)

Traditional SSL algorithms typically employ one-time sample augmentation methods, and the HRC-MS3 method falls into this category as well. To address this limitation, this article introduces an iterative approach to continuously acquire new samples from unknown data, aiming to better leverage

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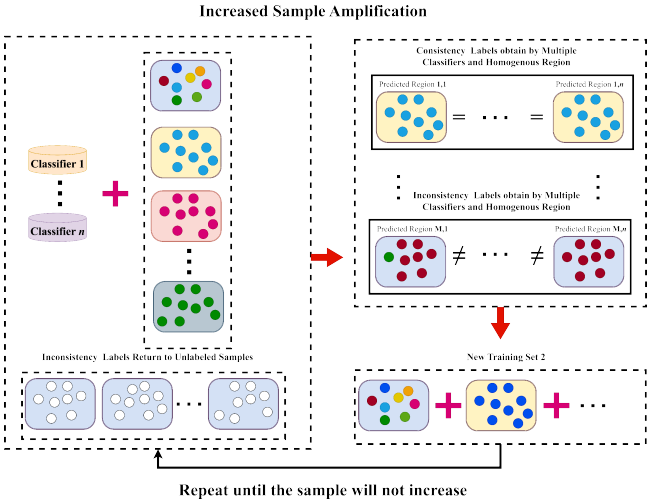


Fig. 2. Flowchart for the proposed IMSE-WDL.

the untapped data. This approach contributes to improving the performance of classification models since more samples can provide additional information for model training. Furthermore, to mitigate the sample augmentation errors that may arise from different-scale SP segmentation, the article introduces a weighted loss function. This loss function is based on constructing a weight matrix from the residuals of different classifiers, which is utilized for multiscale dictionary learning. This approach enables better handling of data at various scales, thereby enhancing classification accuracy. The corresponding flowchart is presented in Fig. 2, and the specific details of this method are as follows.

1) Iterative Sample Expansion: Consider an initial training sample $\boldsymbol{X}^{(0)}$ and testing sample $\boldsymbol{Y}_l^{(0)}$. In each subgraph $\boldsymbol{Y}_l^{(0)}$, the classification results based on three different classifiers can be obtained as follows:

$$\begin{cases} \arg\min\limits_{\boldsymbol{\alpha}^*} \left\| \boldsymbol{Y}_l^{(0)} - \boldsymbol{X}^{(0)}\boldsymbol{\alpha}^* \right\|_2^2 + \lambda \left\| \boldsymbol{\alpha}^* \right\|_2^2 \\ \arg\min\limits_{\boldsymbol{\alpha}^*} \left\| \boldsymbol{\Phi}\left(\boldsymbol{Y}_l^{(0)}\right) - \boldsymbol{\Phi}\left(\boldsymbol{X}^{(0)}\right)\boldsymbol{\alpha}^* \right\|_2^2 + \lambda \left\| \boldsymbol{\alpha}^* \right\|_2^2 \\ \arg\min\limits_{\boldsymbol{\alpha}^*} \left\| \boldsymbol{Y}_l^{(0)} - \boldsymbol{X}^{(0)}\boldsymbol{\alpha}^* \right\|_2^2 + \lambda \left\| \boldsymbol{\alpha}^* \right\|_2^2 \\ \qquad + \dfrac{\eta}{C} \sum\limits_{c=1}^{C} \left\| \boldsymbol{X}^{(0)}\boldsymbol{\alpha}^* - \boldsymbol{X}_c^{(0)}\boldsymbol{\alpha}^*{}_c \right\|_2^2. \end{cases} \tag{19}$$

Similar to HRC-MS3, a new sample augmentation indicator has been proposed. The core idea is to perform multiclassifier EL across different subgraphs, where samples meeting the augmentation criterion are added to the training set, and the remaining samples are utilized to retrain with the augmented training samples

$$\begin{cases} \boldsymbol{X}_l^{(1)} = \boldsymbol{X}^{(0)} \cup \boldsymbol{X}_{\mathbf{1},i}^{(0)}, & \text{if } \boldsymbol{P}_{\mathbf{1},i}^{(0)} = \boldsymbol{P}_{\mathbf{2},i}^{(0)} = \boldsymbol{P}_{\mathbf{3},i}^{(0)} \\ \boldsymbol{Y}_l^{(1)} = \boldsymbol{X}_{\mathbf{1},i}^{(0)}, & \text{else} \end{cases} \tag{20}$$

$$\begin{cases} \boldsymbol{X}^{(\mathbf{1})} = \boldsymbol{X}_{\mathbf{1}}^{(1)} \cup \boldsymbol{X}_{\mathbf{2}}^{(1)} \cup \cdots \cup \boldsymbol{X}_{\boldsymbol{L}}^{(1)} \\ \boldsymbol{Y}^{(\mathbf{1})} = \boldsymbol{Y}_{\mathbf{1}}^{(1)} \cup \boldsymbol{Y}_{\mathbf{2}}^{(1)} \cup \cdots \cup \boldsymbol{Y}_{\boldsymbol{L}}^{(1)} \end{cases} \tag{21}$$

where $\boldsymbol{X}_l^{(1)}$ represents the result of the first round of sample augmentation and $\boldsymbol{P}_{\mathbf{1},i}^{(0)}$, $\boldsymbol{P}_{\mathbf{2},i}^{(0)}$, and $\boldsymbol{P}_{\mathbf{3},i}^{(0)}$ correspond to the classification results of different base classifiers. The process defined by (19) to (21) is repeated $\boldsymbol{t}$ times until the number of training samples no longer increases, resulting in the final training sample set $\boldsymbol{X}^{(t)}$.

2) Multiscale Sample Expansion Results Fusion With WDL: The core idea of the proposed algorithm is based on the homogeneity of land cover types in different subregions obtained through SP segmentation. Therefore, the number of SP blocks is a crucial parameter. Specifically, a smaller number of SP blocks results in larger blocks, leading to smoother segmentation results. On the other hand, a larger number of SP blocks captures more details, potentially generating more complex segments. To overcome the impact of SP segmentation results at different scales on the accuracy of the proposed algorithm, we introduce a multiscale fusion strategy based on WDL. The specific method is as follows.

Assuming that the number of SP blocks is set as $\boldsymbol{L} = \{1, 2, \ldots, Q\}$, the augmented training samples and testing samples obtained based on (19)–(21) are denoted as $\boldsymbol{X}^{(\boldsymbol{t}_1)} = \{\boldsymbol{X}^{(t_1)}, \ldots, \boldsymbol{X}^{(t_Q)}\}, i = 1, \ldots, Q$ and $\boldsymbol{Y}$, respectively. Then, for multiscale segmentation results, different-scale residuals can be obtained according to (12)

$$\boldsymbol{r}_k^{(t_i)} = \left\{ \boldsymbol{r}^{(t_1)}, \ldots, \boldsymbol{r}^{(t_Q)} \right\}, \quad k = 1, \ldots, 3 \tag{22}$$

where $\boldsymbol{r}_k^{(t_i)}$ represents the residual from the kth classifier when the number of subgraphs is l. The corresponding weight matrix is calculated as

$$\boldsymbol{w}_k^{(t_l)} = \frac{e^{-\boldsymbol{r}_k^{(l)}}}{\sum_{k=1}^{K} e^{-\boldsymbol{r}_k^{(t_l)}}} \tag{23}$$

$$\boldsymbol{\alpha}_{(w,k)} = \boldsymbol{w}_k^{(t_l)} \boldsymbol{\alpha}_k \tag{24}$$

where $\boldsymbol{w}_k^{(t_l)}$ represents the sample weights obtained from the kth classifier when the number of subgraphs is l, and $\boldsymbol{\alpha}_{(w,k)}$ is the weighted reconstruction coefficient. The following loss function is proposed according to the reconstruction coefficients

$$\boldsymbol{L} = \arg\min_{\boldsymbol{X}^{(t_l+1)}} \left\| \boldsymbol{Y} - \boldsymbol{X}^{(t_l+1)}\boldsymbol{\alpha}_{(w,k)} \right\|_2^2 + \lambda \left\| \boldsymbol{X}^{(t_l+1)} - \boldsymbol{X}^{(t_l)} \right\|_2^2 \tag{25}$$

where $\|\boldsymbol{Y} - \boldsymbol{X}^{(t_l+1)}\boldsymbol{\alpha}_{(w,k)}\|_2^2$ is used for dictionary reconstruction based on weighted coefficients, $\|\boldsymbol{X}^{(t_l+1)} - \boldsymbol{X}^{(t_l)}\|_2^2$ is used to balance the difference between the reconstructed dictionary and the original training dictionary, and λ is a regularization parameter. The specific solution for this loss function can be written as

$$\frac{\partial \boldsymbol{L}}{\partial \boldsymbol{X}^{(t_l+1)}} = -\left(\boldsymbol{Y} - \partial \boldsymbol{X}^{(t_l+1)}\boldsymbol{\alpha}_{(w,k)}\right)\boldsymbol{\alpha}_{(w,k)}^{\boldsymbol{T}} + \lambda\left(\boldsymbol{X}^{(t_l+1)} - \boldsymbol{X}^{(t_i t_l)}\right) \tag{26}$$

$$\boldsymbol{X}^{(t_l+1)} = \left(\boldsymbol{Y}\boldsymbol{\alpha}_{(w,k)}^{\boldsymbol{T}} + \lambda\right)\left(\boldsymbol{I} + \lambda \boldsymbol{I}\right)^{-1} \tag{27}$$

where $\boldsymbol{X}^{(t_l+1)}$ is the final training sample used to classify the testing sample $\boldsymbol{Y}$. If $\boldsymbol{X}^{(t_l+1)}$ is considered as a new training sample, then the loss functions of the three base classifiers

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TABLE I
SAMPLE DESCRIPTION IN YANGTZE RIVER DELTA

No.	Class	Training	Testing
1	river	2	1271
2	mud flat (muf)	2	474
3	bare land (bal)	2	198
4	impervious surface (ims)	2	1044
5	tree	2	573
6	flood plain (flp)	2	433
7	farmland (far)	2	187
8	plant	2	453
9	lake	2	135
Total		18	4768

can be redefined as follows:

$$\begin{cases} \arg\min_{\boldsymbol{\alpha}^*} \left\| \boldsymbol{Y} - \boldsymbol{X}^{(t_l+1)}\boldsymbol{\alpha}^* \right\|_2^2 + \lambda \left\| \boldsymbol{\alpha}^* \right\|_2^2 \\ \arg\min_{\boldsymbol{\alpha}^*} \left\| \boldsymbol{\Phi}(\boldsymbol{Y}) - \boldsymbol{\Phi}\left(\boldsymbol{X}^{(t_l+1)}\right)\boldsymbol{\alpha}^* \right\|_2^2 + \lambda \left\| \boldsymbol{\alpha}^* \right\|_2^2 \\ \arg\min_{\boldsymbol{\alpha}^*} \left\| \boldsymbol{Y} - \boldsymbol{X}^{(t_l+1)}\boldsymbol{\alpha}^* \right\|_2^2 + \lambda \left\| \boldsymbol{\alpha}^* \right\|_2^2 \\ \qquad + \dfrac{\eta}{C} \sum_{c=1}^{C} \left\| \boldsymbol{X}^{(t_l+1)}\boldsymbol{\alpha}^* - \boldsymbol{X}_c^{(t_l+1)}\boldsymbol{\alpha}_c^* \right\|_2^2. \end{cases} \tag{28}$$

Based on (17) and (18), the ensemble results of classifiers at different scales are as follows:

$$\boldsymbol{P}^{\boldsymbol{l}} = \operatorname{argmax}\left(\boldsymbol{P}^{\boldsymbol{l}}_{\mathbf{1},\boldsymbol{X}^{(t_l+1)}}, \boldsymbol{P}^{\boldsymbol{l}}_{\mathbf{2},\boldsymbol{X}^{(t_l+1)}}, \boldsymbol{P}^{\boldsymbol{l}}_{\mathbf{3},\boldsymbol{X}^{(t_l+1)}} \right) \tag{29}$$

$$\boldsymbol{P}_{\boldsymbol{ms}} = \operatorname{argmax}\left(\boldsymbol{P}^1, \ldots, \boldsymbol{P}^Q \right) \tag{30}$$

where $\boldsymbol{P}^{\boldsymbol{l}}$ represents the fusion result of the multiclassifier when the SP block is the final ensemble result under the multiscale SP segmentation.

IV. EXPERIMENTS

A. Data Description

In this article, three GF-5 Chinese coastland wetland hyperspectral datasets are selected for classification experiments.

The proposed algorithm's effectiveness was verified using the wetland data from the Yangtze River Delta (Nantong) (31°40′–31°55′N, 20°45′–121°10′E) in Jiangsu Province. This wetland, located near a city with a relatively high degree of urbanization, experiences significant impacts from human activities. The dataset contains 268 spectral bands (after removing 62 bad bands) with wavelengths ranging from 0.39 to 2.5 μm. It has a spatial size of 1160 × 1209. The details of the nine classes in this HSI are described in Table I, and the images are shown in Fig. 3.

The second wetland data chosen for the experiment is from Dafeng Natural Reserve (32°56′–33°36′N, 120°42′–120°51′E), located along the sea-shore of the Yellow Sea. Unlike the first wetland data, this area is a nature reserve, characterized by natural biological plants with minimal human impact. The image consists of 256 spectral bands (after removing 74 bad bands) with wavelengths ranging from 0.39 to 2.5 μm. It has a spatial resolution of 30 m and contains 986 × 632 pixels. The descriptions of classes in this data are listed in Table II, and images are shown in Fig. 4.

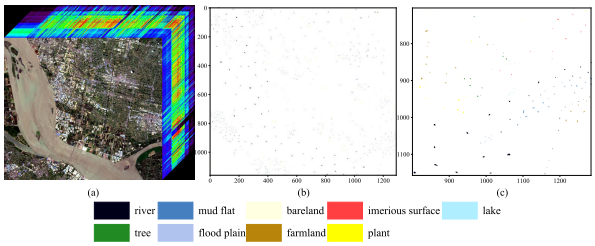


Fig. 3. (a) False-color image, (b) ground truth, and (c) magnified ground truth of the Yangtze River Delta (Nantong) dataset.

TABLE II
SAMPLE DESCRIPTION IN YELLOW RIVER DELTA

No.	Class	Training	Testing
1	acquaculture (acq)	2	391
2	rice	2	188
3	broomcorn (bro)	2	93
4	shallow sea(sha)	2	934
5	suaeda salsa (sus)	2	359
6	aquatic vegetation (aqv)	2	37
7	seep sea (ses)	2	794
8	building (bui)	2	81
9	locust	2	209
10	mud flat (muf)	2	551
11	reed	2	238
12	soybean (soy)	2	108
13	maize	2	94
14	spartina (spa)	2	198
15	river	2	467
16	salt marsh (sam)	2	593
17	intertidal saltwater (ins)	2	452
18	flood plain (flp)	2	66
19	tamarix (tam)	2	131
20	freshwater herbaceous marsh (fhm)	2	70
21	pond	2	375
Total		42	6449

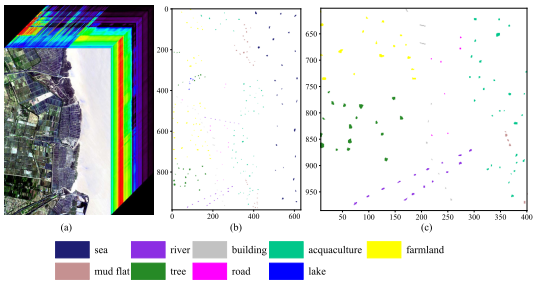


Fig. 4. (a) False-color image, (b) ground truth, and (c) magnified ground truth of the Dafeng Natural Reserve dataset.

The third dataset used to evaluate the proposed methods is the Yellow River Delta dataset [36] (37°15′–38°10′N, 118°10′–119°15′E), with a resolution of 1185 × 1324 pixels. The Yellow River Delta is situated at the estuary of the Yellow River in Huanghe Kou town, Kenli District, Dongying, Shandong province. The spectral bands cover wavelengths ranging from 0.4 to 2.5 μm, with a total of 285 bands remaining after removing 45 bad bands. The image includes

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TABLE III
SAMPLE DESCRIPTION IN DAFENG NATURAL RESERVE

No.	Class	Training	Testing
1	sea	2	1018
2	river	2	274
3	building (bui)	2	87
4	aquaculture (acq)	2	744
5	farmland (far)	2	916
6	mud flat (muf)	2	526
7	tree	2	722
8	road	2	148
9	lake	2	77
Total		18	4512

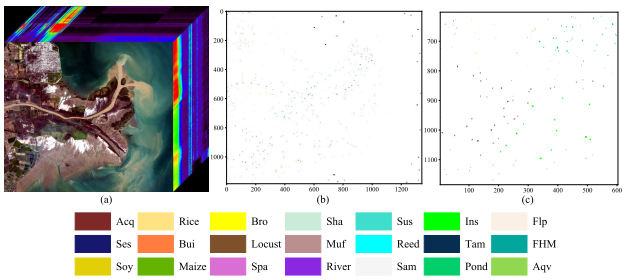


Fig. 5. (a) False-color image, (b) ground truth, and (c) magnified ground truth of the Yellow River Delta dataset.

21 classes, and the spatial resolution is 30 m. Detailed information for each class is described in Table III, and images of this dataset are shown in Fig. 5.

B. Experimental Setup

1) Initial Training Size: For HSI classification, the issue of limited sample size has always been a challenging task. Existing research often presents a wide range of definitions for few-shot size in the context of HSI classification, ranging from 1%–5% of the total sample quantity. Given the absence of a precise definition for few-shot sizes, this study employs just two samples per class for training. Specifically, for the Yangtze River Delta, Jiangsu Dafeng Natural Reserve, and Yellow River Delta datasets, only 0.4%, 0.6%, and 0.4% of the samples were used for training, with the remaining samples reserved for testing. Additionally, experiments were conducted with one to five samples per class for training, ensuring the proposed method's ability to generalize across different initial sample sizes. The sample description is listed in Tables I–III.

2) Parameter Setting: For the algorithms proposed in the article, there are two key parameters to be considered: the number of SPs and the regularization parameter λ for dictionary learning. The specific details are as follows. The parameter SP is selected from {1000, 1500, 2000, 2500}. The regularization parameter λ is selected from {0.1, 0.2, 0.3, 0.4, 0.5}. The assessment of classification performance for both the comparative methods and the proposed techniques in this section relies on metrics such as overall accuracy (OA), average accuracy (AA), and Kappa indexes.

C. Comparative Algorithms

To comprehensively validate the two algorithms proposed in the article, a variety of classical and state-of-the-art machine learning and deep learning classification algorithms were employed as comparative methods. The specific details are as follows.

1) Ensemble Algorithms:

a) Light gradient boosting machine (LightGBM) [37]: LightGBM is an open-source, distributed, and high-performance gradient-boosting framework designed for machine learning tasks, particularly for tasks involving tabular data, such as classification, regression, and ranking.

2) Deep Learning Algorithms:

a) Graph convolutional network (GCN): GCN is a deep learning model used for processing graph-structured data, capturing relationships and features between nodes through convolution operations, generating new node representations through message passing and nonlinear transformations, and widely applied to graph data analysis tasks such as node classification and graph classification.

b) Graph attention network (GAT): GAT is a graph neural network model used for machine learning tasks involving graph data or graph-structured data. The core idea of GAT is to learn relationships and features between nodes using a self-attention mechanism without the need for fixed convolutional operations. This allows GAT to dynamically assign different weights to each node, thereby better capturing important relationships between nodes.

c) Graph sample and aggregation (GraphSAGE) [38]: GraphSAGE is a graph neural network model for graph data, designed to learn low-dimensional representations of nodes for various graph analysis tasks. The core idea of GraphSAGE is to generate node representations by sampling neighboring nodes and aggregating their information, capturing both local and global structural information of nodes. The method proposed in the article is essentially a semi-supervised sample augmentation algorithm. Therefore, to ensure fairness, we have combined all the comparative algorithms with the LP [39], which is an SSL algorithm commonly applied in tasks such as social network analysis, graph data mining, and community detection.

Its core idea is to propagate label information on the graph structure for classification or clustering purposes. The samples expanded through the LP algorithm are used to enhance the following comparative algorithms: LP-LightGBM, LP-GCN, LP-GAT, and LP-GraphSAGE.

Furthermore, the tri-training [40] algorithm is also combined with LightGBM [41] and CR-based classifiers used in our proposed algorithm to demonstrate the superior performance of our proposed method in SSL (Tri1 is tri-training based on CatBoost [37], LightGBM, and GBDT; Tri2 is tri-training with CR-based classifiers).

D. Classification Performance and Maps

In this study, classification of three sets of wetland data in the Yangtze River Delta, Yellow River Delta, and Dafeng Natural Reserve was performed. The performance of

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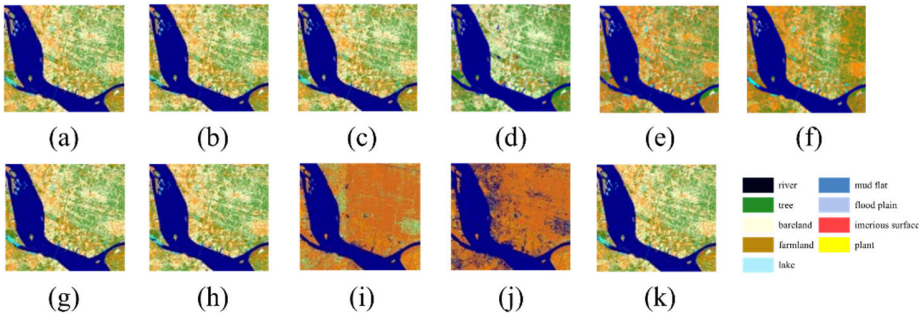


Fig. 6. Classification maps obtained by different models of the Yangtze River Delta (Nantong). (a)–(k) Correspond to LP-LightGBM, LP-GCN, LP-GAT, LP-GraphSAGE, Tr1, Tr2, M1-1000, M1-1500, M1-2000, M1-2500, and M2.

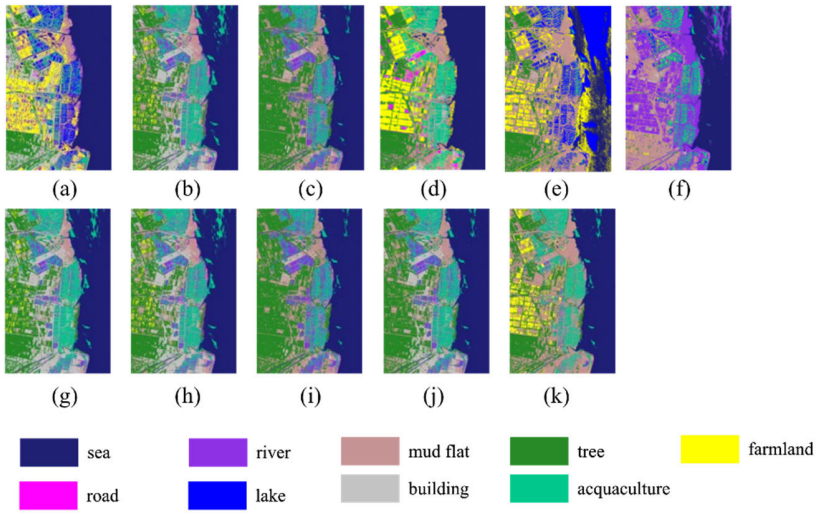


Fig. 7. Classification maps obtained by different models of the Dafeng Natural Reserve. (a)–(k) Correspond to LP-LightGBM, LP-GCN, LP-GAT, LP-GraphSAGE, Tr1, Tr2, M1-1000, M1-1500, M1-2000, M1-2500, and M2.

various classification methods was compared, including LP-LightGBM, LP-GCN, LP-GAT, LP-GraphSAGE, as well as HRC-MS3 with different parameter settings (M1-1000, M1-1500, M1-2000, M1-2500), and the improved IMSE-WDL (M2). The specific analysis is as follows.

1) Results for Yangtze River Delta: From Table IV and Fig. 6, regarding the Yangtze River Delta data, first, from the perspective of OA, it was observed that good accuracy results were achieved by different parameter settings of HRC-MS3. Especially noteworthy was M1-2000, where an accuracy of 84.69% was attained, which was comparable to other benchmark algorithms. Meanwhile, IMSE-WDL was found to exhibit outstanding performance, with an accuracy of 82.07%, surpassing all comparative algorithms significantly. It was concluded that HRC-MS3 can compete with benchmark algorithms under certain parameter settings, and IMSE-WDL demonstrates a significant advantage in OA.

In terms of AA, different parameter settings of HRC-MS3 still displayed a commendable performance. Of particular note was M1-2500, with an AA of 78.91%, surpassing some benchmark algorithms. Meanwhile, IMSE-WDL continued to perform the best in this metric, with an AA of 79.50%. It again confirms the potential of HRC-MS3 and the excellence of IMSE-WDL in overall performance. Additionally, the Kappa coefficient, an important indicator for assessing classification performance, was evaluated. The results showed that Kappa coefficients higher than 0.75 were achieved by different parameter settings of HRC-MS3 and IMSE-WDL, indicating their outstanding performance. The Kappa coefficient for IMSE-WDL was 0.80, significantly higher than other methods. It was noted that HRC-MS3 also exhibited relatively high Kappa coefficients.

Most importantly, the superiority of IMSE-WDL is not only evident in classification accuracy but also in its independence from manual parameter tuning. In contrast, HRC-MS3 is sensitive to parameter setting, requiring a considerable amount of time and effort for parameter adjustments. IMSE-WDL, on the other hand, can achieve optimal performance without the need for careful parameter tuning.

2) Results for Yellow River Delta: From Table V and Fig. 7, first, regarding OA, we can observe that IMSE-WDL(M2) stands out, achieving an impressive accuracy of 86.87% on the Yellow River Delta dataset. This performance is significantly better than all the comparative algorithms. Different parameter settings for HRC-MS3 also exhibit commendable performance to some extent, especially M1-1500, which attains an accuracy of 86.36%. This indicates that both HRC-MS3 and IMSE-WDL have advantages in terms of overall classification accuracy.

Furthermore, when considering the results for AA, IMSE-WDL continues to perform the best, reaching 85.25%, surpassing all comparative algorithms and various param-

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TABLE IV

PERFORMANCE OF DIFFERENT METHODS FOR THE YANGTZE RIVER DELTA (0.4% TOTAL SAMPLES AS TRAINING SET; LP-LIGHT AND LP-GRAPH IS LP-LIGHTGBM AND LP-GRAPHSAGE)

class	LP-Light	LP-GCN	LP-GAT	LP-Grap	Tri1	Tri2	M1-1000	M1-1500	M1-2000	M1-2500	**M2**
1	96.88±3.46	96.67±4.48	96.75±3.78	97.22±3.72	99.48±0.91	98.19±1.57	97.59± 1.07	97.46±0.99	97.19±1.54	97.09±1.24	97.48±1.19
2	70.12±20.79	84.00±14.77	79.41±16.64	73.66±23.39	56.43±47.43	51.25±24.52	57.82± 41.71	56.02± 41.08	57.01± 42.25	56.14± 39.59	54.92± 46.64
3	80.40±10.21	70.74±13.77	80.12±19.75	86.79±11.09	97.64±2.10	64.31±7.83	73.40± 21.77	78.96± 14.83	81.99± 12.55	80.98± 12.98	74.24± 18.71
4	89.22±6.88	88.13±10.62	87.44±12.04	86.73±11.69	93.05±2.33	59.54±7.51	94.83± 4.79	92.69± 8.69	94.12± 4.75	95.72± 2.67	94.83 ±3.04
5	70.18±9.97	73.49±12.45	72.08±9.40	66.90±10.70	26.17±23.24	49.88±24.68	68.15± 11.64	58.02± 11.53	51.60± 11.32	59.01± 13.75	60.49 ±11.14
6	41.65±15.88	19.69±10.01	27.26±16.41	40.47±25.89	54.50±46.79	30.66±11.83	58.79± 37.13	58.09± 31.33	60.34± 35.61	60.55± 31.34	58.86± 41.06
7	90.15±8.87	90.93±9.60	91.55±8.49	89.87±9.90	78.44±18.77	32.18±19.05	90.15± 11.41	91.84± 10.30	90.38± 9.48	88.30± 12.65	90.69 ±9.16
8	65.31±17.96	74.99±10.28	70.46±22.12	61.70±25.23	44.38±16.91	55.49±19.50	81.35± 10.74	80.40± 11.21	80.72± 10.14	80.78± 9.87	89.56 ±5.20
9	85.97±10.35	86.50±13.32	87.23±12.58	87.97±14.39	99.78±0.22	77.34±19.18	94.26± 7.65	93.23± 4.60	93.38± 4.56	91.61± 7.11	94.48 ±5.74
OA	76.77±2.24	78.51±1.73	78.69±2.80	77.05±4.23	73.48±3.80	64.22±4.52	82.46± 2.09	81.61± 2.47	82.07± 2.32	81.84± 1.92	**83.78 ±1.40**
AA	76.36±1.96	76.43±1.25	77.32±2.92	76.46±2.76	72.21±4.41	57.65±4.39	79.60± 1.85	78.41± 2.63	78.52± 2.93	78.91± 2.67	**79.50 ±2.66**
Kappa	0.73±0.02	0.74±0.02	0.75±0.03	0.73±0.05	0.69±0.04	0.57±0.05	0.79±0.03	0.78 ±0.03	0.79± 0.03	0.78 ±0.02	**0.80± 0.02**

TABLE V

PERFORMANCE OF DIFFERENT METHODS FOR THE YELLOW RIVER DELTA (0.6% TOTAL SAMPLES AS TRAINING SET; LP-LIGHT AND LP-GRAPH IS LP-LIGHTGBM AND LP-GRAPHSAGE)

class	LP-Light	LP-GCN	LP-GAT	LP-Graph	Tr1	Tr2	M1-1000	M1-1500	M1-2000	M1-2500	**M2**
1	86.79±5.26	87.55±5.27	90.54±2.30	88.23±6.05	53.28±40.83	77.15±1.26	63.26±8.39	63.17±7.76	57.46 ±0.90	61.13± 7.53	55.67 ±7.22
2	95.05±8.58	94.88±8.87	94.88±8.87	94.88±8.87	91.77±9.76	80.10±20.78	93.91± 8.16	93.07± 7.79	92.11± 7.69	96.64± 2.91	94.58 ±8.63
3	93.52±2.45	98.77±0.53	93.52±5.63	95.68±1.07	98.77±0.53	75.93±20.05	97.53± 1.41	97.53± 0.53	97.53± 1.93	95.99± 1.07	98.46± 1.07
4	74.47±13.33	82.45±5.13	57.09±49.52	49.47±32.58	82.09±10.75	62.77±23.01	88.30± 0.00	88.30±0.00	87.77± 0.92	87.94 ±0.61	88.30± 0.00
5	39.92±15.63	34.98±13.20	20.99±5.66	23.05±6.33	34.16±0.71	31.69±25.43	52.67 ±29.01	42.80± 26.40	40.33± 14.83	29.22 ±14.26	45.27± 19.38
6	63.47±40.32	65.60±51.51	62.76±54.38	51.42±47.86	86.88±22.73	73.05±30.25	79.43 ±33.80	69.86± 50.38	69.86± 50.38	71.99± 46.68	71.99± 46.69
7	50.90±20.58	53.05±34.50	0.00±0.00	19.71±31.39	92.83±3.78	50.18±30.89	71.68± 24.05	57.71± 27.29	70.61± 29.49	73.84 ±24.83	71.68 ±24.05
8	99.84±0.28	100.00±0.00	68.58±54.42	100.00±0.00	100.00±0.00	95.37±3.53	100.00±0.00	100.00± 0.00	100.00± 0.00	100.00± 0.00	100.00 ±0.00
9	94.61±3.43	97.98±2.02	98.65±1.91	98.48±1.52	100.00±0.00	87.37±10.96	96.30± 3.25	97.98± 3.50	98.65± 2.33	99.16 ±1.46	98.32 ±2.49
10	59.89±10.97	66.49±22.71	72.23±23.56	75.30±26.55	55.42±46.48	50.14±22.03	61.56±14.02	67.52± 22.72	67.59± 23.90	55.78 ±6.31	66.85± 23.03
11	95.95±2.60	96.98±2.81	99.94±0.10	99.94±0.10	97.16±2.12	71.26±37.84	98.97±1.63	99.94± 0.10	98.97± 1.63	99.94 ±0.10	98.97 ±1.63
12	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	66.02±29.92	100.00±0.00	100.00± 0.00	100.00± 0.00	100.00 ±0.00	100.00 ±0.00
13	89.42±4.58	98.51±2.57	98.14±3.22	100.00±0.00	100.00±0.00	96.56±3.58	99.91± 0.16	100.00± 0.00	100.00± 0.00	100.00 ±0.00	100.00± 0.00
14	67.65±12.01	80.11±5.59	80.67±11.30	54.62±37.75	32.49±9.86	68.07±7.36	66.67± 10.55	63.17 ±15.64	68.35± 12.64	77.03 ±10.67	70.03± 13.36
15	96.85±3.30	96.12±0.77	97.41±0.54	98.59±0.10	99.10±1.56	100.00±0.00	100.00± 0.00	100.00± 0.00	100.00± 0.00	100.00 ±0.00	100.00 ±0.00
16	80.24±15.40	81.42±15.73	93.29±6.34	91.59±14.56	95.72±6.12	73.38±20.15	95.72 ±6.12	92.63±12.77	92.70 ±12.65	97.79 ±1.92	92.85± 8.68
17	67.94±40.13	69.72±39.96	49.62±47.82	67.68±24.35	81.68±11.01	79.90±4.85	73.03 ±24.26	89.82± 10.36	89.31 ±9.53	93.89± 2.64	89.31± 10.27
18	57.69±9.09	53.42±17.19	33.60±36.29	31.73±46.28	72.09±25.07	50.93±29.44	88.98± 12.29	82.76± 15.39	83.82± 14.05	82.58 ±15.09	88.27 ±10.92
19	92.93±0.88	92.42±4.54	42.93±37.37	68.18±43.47	99.49±0.87	79.29±7.16	92.93± 1.75	92.42± 1.52	93.43 ±0.87	91.41 ±0.87	92.42 ±1.52
20	56.19±29.98	60.00±30.00	33.33±14.31	43.81±36.54	0.00±0.00	78.57±16.48	71.43± 37.14	62.38± 40.26	57.62± 39.55	62.38 ±23.92	70.95 ±45.45
21	96.40±1.56	97.30±0.00	96.40±1.56	97.30±0.00	97.30±0.00	81.98±12.19	95.50± 1.56	96.40±1.56	96.40± 1.56	95.50 ±1.56	96.40 ±1.56
OA	82.22±3.13	79.54±6.99	81.58±3.94	74.76±11.15	81.76±3.81	72.90±4.84	86.39± 0.81	86.36 ±1.63	86.19 ±1.76	86.59 ±1.07	**86.87± 2.33**
AA	79.03±4.21	73.24±10.91	70.70±6.19	68.27±10.53	79.54±1.24	72.84±6.35	85.13± 1.65	83.69 ±2.48	83.93± 2.25	83.67 ±1.12	**85.25± 2.53**
Kappa	0.81±0.03	0.78±0.08	0.80±0.04	0.74±0.12	0.81±0.04	0.71±0.05	0.85 ±0.01	0.85 ±0.02	0.85± 0.22	0.85± 0.02	**0.86 ±0.03**

eter combinations of HRC-MS3. However, M1-2500 also demonstrates a relatively high AA, at 83.67%, which is close to IMSE-WDL. This emphasizes that HRC-MS3 can compete with IMSE-WDL under certain parameter settings and excel in overall performance.

As for the Kappa coefficient, the results reveal that IMSE-WDL has a Kappa coefficient of 0.86, higher than other methods, including comparative algorithms and different parameter settings of HRC-MS3. Simultaneously, the Kappa coefficient for M1-2500 is also relatively high, at 0.85. This suggests that both HRC-MS3 and IMSE-WDL exhibit outstanding performance, with M1-2500 capable of achieving performance levels comparable to IMSE-WDL in certain scenarios.

It is worth emphasizing that the advantage of IMSE-WDL lies in its ability to operate without the need for manual parameter tuning, while HRC-MS3 exhibits performance variations under different parameter settings, requiring parameter adjustments. This makes IMSE-WDL a convenient and high-performance classification method, whereas HRC-MS3 excels when fine-tuning of parameters is needed.

3) Results for Dafeng Natura Reserve: From Table VI and Fig. 8, for the Dafeng Natural Reserve dataset, from the perspective of OA, IMSE-WDL (M2) exhibits outstanding

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TABLE VI
PERFORMANCE OF DIFFERENT METHODS FOR THE DAFENG NATURA RESERVE (0.4% TOTAL SAMPLES AS TRAINING SET; LP-LIGHT AND LP-GRAPH IS LP-LIGHTGBM AND LP-GRAPHSAGE)

class	LP-Light	LP-GCN	LP-GAT	LP-Graph	Tr1	Tr2	M1-1000	M1-1500	M1-2000	M1-2500	M2
1	98.14±3.23	98.64±2.36	98.61±2.41	97.98±3.50	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00
2	73.44± 4.89	82.50±10.82	81.70±12.13	67.65±11.88	79.40±0.88	63.94±4.22	80.29±0.72	80.61±0.19	80.54±0.29	80.67± 0.22	80.74±0.29
3	68.05±9.50	63.70 ±20.11	73.50±13.33	64.69±23.06	75.45±3.19	75.45±10.92	81.76±18.66	79.05±4.87	83.78±16.38	88.74±7.03	83.33± 18.00
4	95.70± 6.48	97.33 ±4.63	96.59±4.95	96.94±4.34	99.23±1.33	79.31±13.94	99.62±0.66	100.00±0.00	99.62± 0.66	99.62± 0.66	99.62±0.66
5	73.45±18.72	64.20±35.64	70.21±26.51	82.64±7.66	79.59±10.89	64.41±8.24	82.13±9.43	82.82±8.96	78.93±6.64	81.37± 9.68	84.90± 6.06
6	43.80±23.49	38.65± 39.36	34.94±35.36	38.52±17.70	17.15±3.18	32.85±17.76	55.96±29.92	57.42±31.95	50.49±34.70	56.69± 64.01	54.99± 33.87
7	81.54±7.91	82.05± 9.88	81.68±7.58	84.10±5.90	95.11±2.52	59.65±18.33	90.72±3.85	91.41±3.85	90.63±3.81	90.90± 3.44	91.51± 3.46
8	76.62±4.58	78.29± 7.55	79.61±3.88	81.18±12.87	62.01±12.05	51.03±17.57	89.38±9.46	87.01±6.20	85.75±6.66	83.02± 10.60	90.95±6.86
9	88.04±6.52	85.04 ±10.81	90.93±1.33	88.62±7.06	92.21±9.00	67.53±14.29	79.65±8.65	78.35±18.65	81.82±9.80	86.15± 16.24	83.12± 13.50
OA	79.29±1.73	78.96±4.63	79.76±2.57	81.23±2.41	80.42±0.94	68.20±5.40	87.21±0.16	87.09±1.50	85.75± 1.18	86.47±0.67	**88.26±1.26**
AA	77.64±2.13	76.41±0.92	78.24±2.34	78.39±2.19	77.79±1.12	66.02±5.03	84.39±1.22	84.08±2.97	83.51± 1.86	85.24±1.65	**85.46±2.41**
Kappa	0.75±0.02	0.75±0.05	0.76±0.03	0.78±0.03	0.77±0.01	0.62±0.07	0.85±0.00	0.85±0.02	0.83± 0.02	0.84±0.01	**0.86± 0.02**

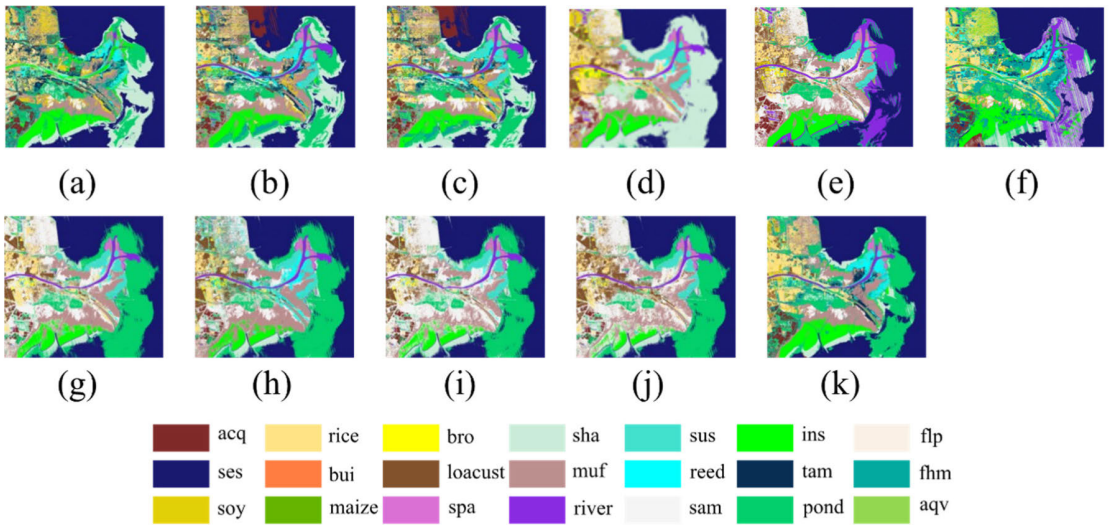


Fig. 8. Classification maps obtained by different models of the Yellow River Delta. (a)–(k) Correspond to LP-LightGBM, LP-GCN, LP-GAT, LP-GraphSAGE, Tr1, Tr2, M1-1000, M1-1500, M2-2000, M1-2500, and M2.

performance, achieving an accuracy rate of 88.26%, markedly superior to all comparative algorithms. HRC-MS3, under different parameter settings, also demonstrates commendable performance on this dataset, particularly M1-1000, M1-1500, M1-2000, and M1-2500, which achieve accuracy rates of 87.21, 87.09, 85.75, and 86.47, respectively. This indicates that both HRC-MS3 and IMSE-WDL possess advantages in overall classification accuracy.

Upon further examination of the AA results, IMSE-WDL continues to outperform, reaching 85.46%, surpassing all comparative algorithms and various parameter combinations of HRC-MS3. Simultaneously, M1-1500 exhibits relatively high AA at 84.08%, in close proximity to IMSE-WDL. This underscores HRC-MS3's ability to compete with IMSE-WDL under certain parameter settings and excel in terms of overall performance.

In addition, IMSE-WDL's Kappa coefficient is 0.86, higher than other methods, including comparative algorithms and various parameter settings of HRC-MS3. Meanwhile, M1-2500's Kappa coefficient is also relatively high, at 0.84. This suggests that both HRC-MS3 and IMSE-WDL perform admirably in terms of model performance, with M1-2500 achieving performance levels comparable to IMSE-WDL in certain scenarios.

It should be emphasized that IMSE-WDL's advantage lies in not requiring manual parameter tuning, whereas HRC-MS3 exhibits performance variations under different parameter settings, necessitating parameter adjustments. This makes IMSE-WDL a convenient and high-performance classification method, while HRC-MS3, with the need for fine-tuning parameters, can achieve outstanding performance.

V. DISCUSSION

A. Analysis of Parameter Sensitivity

1) Influence of Different Initial Training Sizes for HRC-MS3: From Fig. 9(a), for the Yangtze River Delta dataset, it can observe changes in OA with different training sizes. As the number of samples for each class increases, the classification accuracy improves significantly. This indicates

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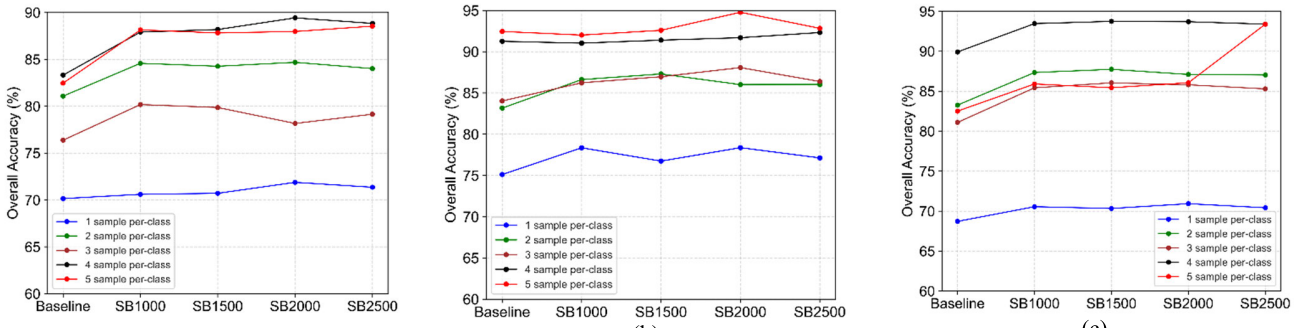


Fig. 9. Classification performance with different initial training sizes and parameter SB. (a) Yangtze River Delta. (b) Yellow River Delta. (c) Jiangsu Dafeng Natural Reserve.

that having more training samples is highly beneficial for this dataset.

Similarly, in Fig. 9(b), for the Dafeng Natural Reserve dataset, OA also significantly improves with an increase in training sizes. This improvement is particularly noteworthy, highlighting that augmenting the number of training samples has a pronounced impact on performance.

In the case of the Yellow River Delta dataset, again, increasing the training size leads to an improvement in OA. Fig. 9(c) also shows that having more training data is highly effective in enhancing the performance of this dataset.

2) Influence of Parameter SB for HRC-MS3: From Fig. 9(a), for the Yangtze River Delta dataset, it can be observed that under different values of the superpixel block (SB) parameter, the improved algorithm consistently outperforms the baseline algorithm in terms of OA. Even when SB increases from 1000 to 2500, OA remains higher than the performance level of the baseline algorithm. This indicates that the improved algorithm consistently exhibits better classification performance compared to the baseline algorithm across various SB parameter settings, demonstrating a higher level of classification accuracy.

Similarly, in Fig. 9(b), for the Dafeng Natural Reserve dataset, the improved algorithm outperforms the baseline algorithm under different SB parameter settings. Whether it is at SB = 1000 or SB = 2500, the OA of the improved algorithm consistently surpasses that of the baseline algorithm. This further validates the effectiveness of the improved algorithm.

From Fig. 9(c), in the case of the Yellow River Delta dataset, the improved algorithm also demonstrates superior performance under various SB parameter settings. As SB parameters increase from 1000 to 2500, the OA of the improved algorithm continues to outperform the baseline algorithm, highlighting the robustness and performance enhancement of the improved algorithm relative to the baseline.

Through this parameter analysis, it can be concluded that the improved algorithm not only demonstrates performance improvement under different training size settings but is also consistently outperformed by the baseline algorithm across various SB parameter settings.

In summary, for the Yangtze River Delta, Dafeng Nature Reserve, and Yellow River Delta datasets, the OA significantly improved with the increase in the number of training samples.

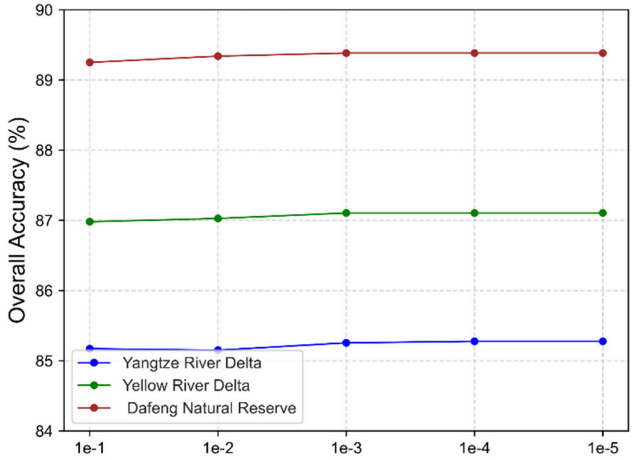


Fig. 10. Classification performance with different λ.

This indicates that for these datasets, the setting of 1–5 samples per class as training samples in this article can be considered as "small samples." Meanwhile, under different SB parameter values, the improved algorithm showed higher classification accuracy than the baseline algorithm on all datasets. Even when the SB parameter increased from 1000 to 2500, the improved algorithm's OA was still higher than that of the baseline algorithm. These results suggest that for hyperspectral remote sensing image classification in supervised and SSL, the definition of "small samples" should start from the case with the fewest samples, that is, each class only selects one sample as training data. Even under this condition, the improved algorithm still demonstrated excellent performance, indicating that it has good classification capabilities even with an extremely limited number of samples. Therefore, it can be considered that this setting of one sample per class is an effective "small sample" threshold.

B. Influence of Different Parameter λ for IMSE-WDL

Fig. 10 presents the effect of various values for initial training size and parameter λ on performance for IMSE-WDL in three wetland datasets.

For the Yangtze River Delta dataset, as λ decreases from 1e-1 to 1e-5, the classification accuracy remains relatively

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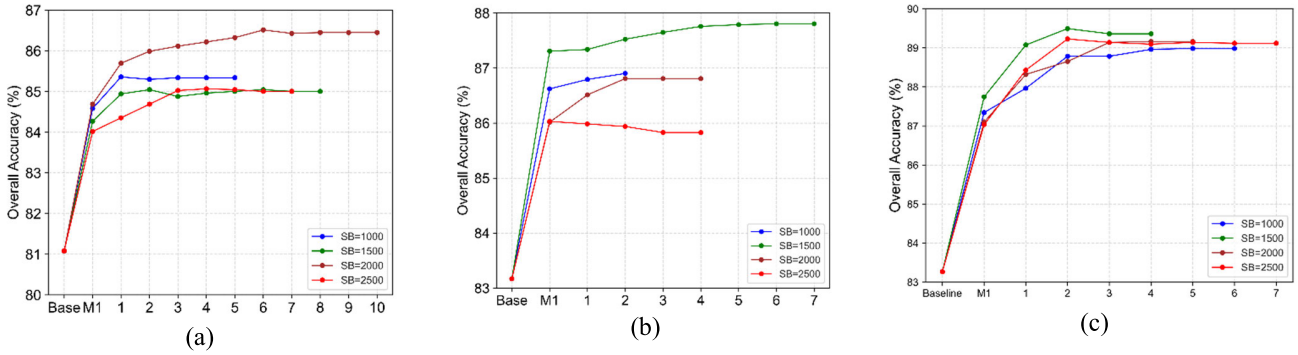


Fig. 11. Classification performance with different parameter SB on the baseline, HRC-MS3, and iterative sample selection process. (a) Yangtze River Delta. (b) Yellow River Delta. (c) Jiangsu Dafeng Natural Reserve.

stable, ranging from approximately 87%. This indicates that the variation in the regularization parameter λ has a relatively small impact on the performance on this dataset.

Similarly, for the Dafeng Natural Reserve dataset, as λ decreases from 1e-1 to 1e-5, OA remains within a relatively stable range, approximately between 89% and 90%. This suggests that, like the previous dataset, the change in the regularization parameter λ has a relatively minor impact on performance. The model's performance remains stable across different levels of regularization.

For the Yellow River Delta dataset, as λ decreases, OA also remains within a relatively stable range, approximately between 85% and 86%. Similar to the first two datasets, this indicates that the variation in the regularization parameter λ has a limited effect on performance. The model's performance remains relatively stable across different regularization strengths.

Overall, for these three datasets, the impact of λ on classification accuracy is relatively small. The models exhibit stable performance under different λ settings. This may suggest that these datasets have some robustness to a certain degree of regularization, allowing the model to maintain good performance under varying levels of regularization strength. In practical applications, an appropriate λ value can be selected to balance model complexity and performance without overfitting or underfitting the data.

C. Effectiveness Analysis of Sample Selection Method

1) Effectiveness Analysis of Iterative Sample Selection Process: Fig. 11(a)–(c) illustrates the trend of OA with the number of iterations in sample selection under various SB settings.

a) Yangtze River Delta dataset: From Fig. 11(a), for SB = 1000, the line chart shows that the OA starts from a relatively low initial point and gradually rises with the increase of iterations, tending to stabilize in the later iterations. This demonstrates that in the initial iterations, the iterative sample selection process has significantly improved the model's performance. For SB = 1500, the line chart also starts from a relatively low point but quickly rises and stabilizes in the subsequent iterations, underscoring the advantage of the iterative sample selection process. For SB = 2000, the line chart reveals a steady growth in OA, indicating that with a larger SB value, the iterative process can further enhance performance. For SB = 2500, although the growth in OA seems a bit slower, there is a noticeable performance improvement after the initial iterations.

b) Dafeng Natural Reserve: From Fig. 11(b), for SB = 1000, the line chart gradually rises from an initial point and tends to stabilize toward the end, showcasing the potency of the iterative sample selection process in this dataset. For SB = 1500, the rise in OA is more pronounced, highlighting the strong advantage of the iterative sample selection process in this particular SB setting. For SB = 2000, the line chart displays a steady rise in OA, starting from a lower point. For SB = 2500, the line trend demonstrates a steady growth in OA, reaffirming the method's efficacy.

c) Yellow River Delta dataset: From Fig. 11(c), for SB = 1000, the line chart depicts a steady rise from a starting point to a higher one. For SB = 1500, the growth trend in OA is evident. For SB = 2000, the line chart unveils a steady growth in OA. For SB = 2500, the line chart presents an initial rise followed by a slight downward trend, which may suggest that in certain scenarios, more iterations may not always yield better results.

In conclusion, the line charts showcase that in most datasets and SB settings, the iterative sample selection process continuously optimizes and improves model performance. However, an evident issue is the decrease in classification accuracy caused by excessive iterations in certain scenarios. To address this, the multiscale WDL module was introduced into the iterative sample selection process, and as a result, IMSE-WDL was formed.

2) Distribution Analysis of Initial Training Set, Amplification Samples, and Testing Set for IMSE-WDL: To validate the effectiveness and reliability of the proposed SSL method, t-SNE algorithm was employed in the article to visualize the distributions of augmented samples and the testing samples. Fig. 12(a) illustrates the distribution of the original training samples. Subsequently, Fig. 12(b)–(e) displays the results of sample augmentation using IMSE-WDL with different SB parameters without the final dictionary learning integration. Finally, Fig. 12(f) depicts the distribution of testing samples that need to be classified.

For the Yangtze River Delta dataset, as observed in Fig. 12(b)–(e), when compared to Fig. 12(f), it is obvious

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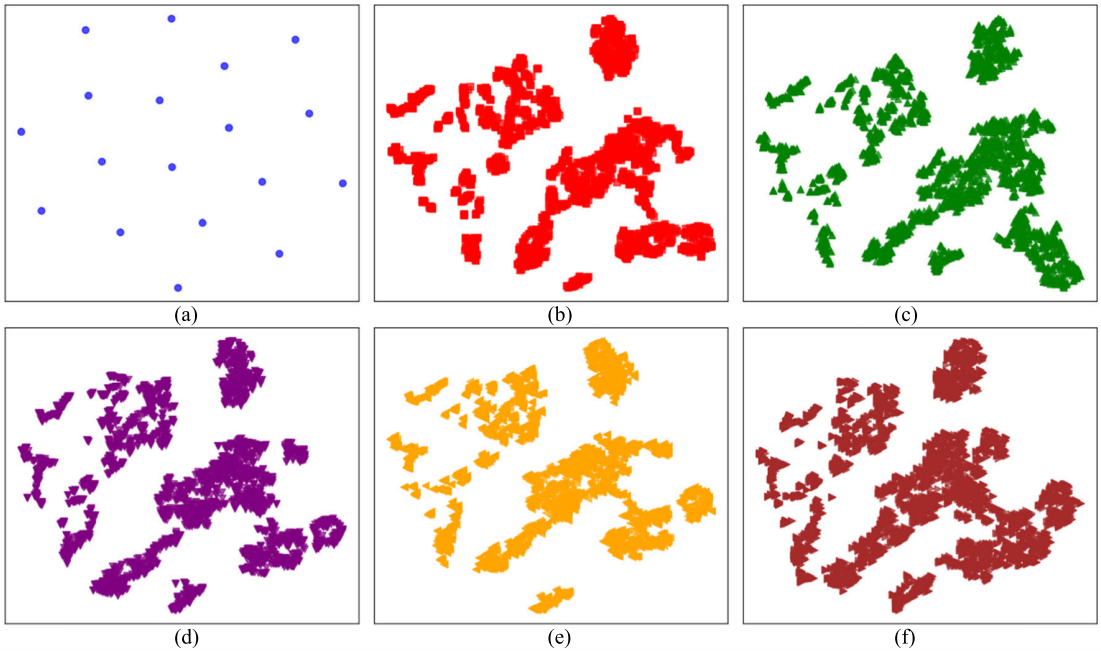


Fig. 12. Visualizing the distribution of selective samples and testing set (Yangtze River Delta). (a) Training data. (b) Sample selection by SB = 1000. (c) Sample selection by SB = 1500. (d) Sample selection by SB = 2000. (e) Sample selection by SB = 2500. (f) Testing samples.

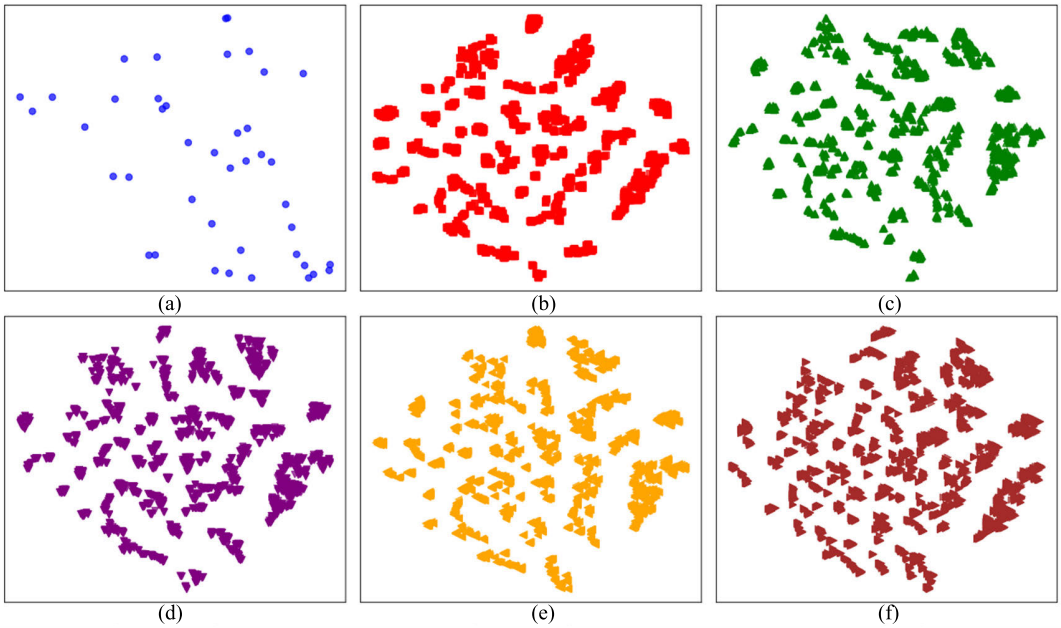


Fig. 13. Visualizing the distribution of selective samples and testing set (Yellow River Delta). The performance remains stable under different regularization strengths. (a) Training data. (b) Sample selection by SB = 1000. (c) Sample selection by SB = 1500. (d) Sample selection by SB = 2000. (e) Sample selection by SB = 2500. (f) Testing samples.

the augmented samples obtained through the proposed IMSE-WDL in the article exhibit a high degree of similarity to the distribution of testing samples, regardless of the parameter settings. The same analysis approach applies to the Yellow River Delta and Dafeng Natural Reserve datasets. Similar results can be inferred from Figs. 13 and 14.

In summary, these results indicate the effectiveness of the augmentation method. It is suggested that the augmentation method, to some extent, is capable of preserving data consistency, as it generates samples with distributions similar to the testing dataset without introducing significant biases or altering the data distribution. The results also imply that the model may have improved generalization performance in this scenario. Since the distribution of augmented samples aligns with the distribution of testing samples, the model is more likely to perform well on the testing samples after training on the augmented samples. This contributes to enhancing the model's ability to generalize to unseen data. Last, the

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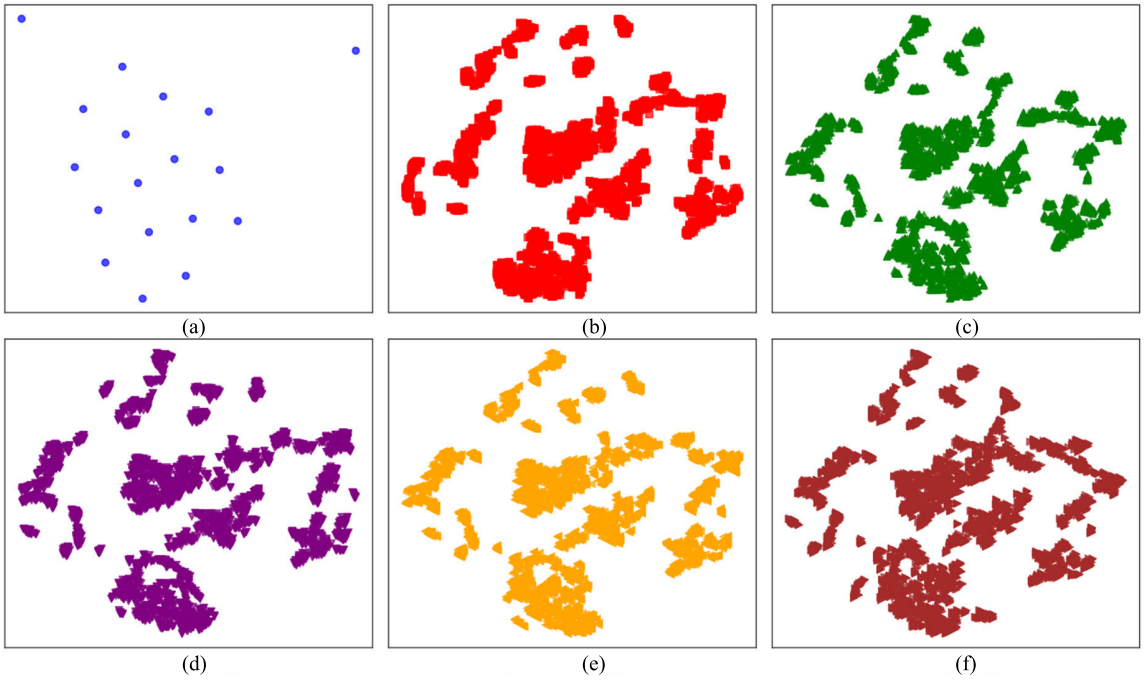


Fig. 14. Visualizing the distribution of selective samples and testing set (Dafeng Natural Reserve). (a) Training data. (b) Sample selection by SB = 1000. (c) Sample selection by SB = 1500. (d) Sample selection by SB = 2000. (e) Sample selection by SB = 2500. (f) Testing samples.

TABLE VII
CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR PAVIA UNIVERSITY (FIVE SAMPLES PER CLASS)

	LP-SVM	LP-CRC	LP-ProCRC	LP-Light	LP-GCN	LP-Graph	M1-1000	M1-1500	M1-2000	M1-2500
OA	65.98	43.49	60.46	53.27	47.65	47.79	**71.49**	64.59	71.06	70.11
AA	0.33	0.42	57.6	68.56	63.91	65.13	**67.71**	62.94	67.41	61.03
Kappa	40.63	56.58	0.47	0.44	0.38	0.39	**0.62**	0.53	0.61	0.59

consistency in distribution may also indicate that the augmentation method has not introduced noticeable errors or abnormal samples. This helps in maintaining data quality and stability, reducing adverse effects on the model.

D. Applicability Analysis for Multisource HSI Datasets

To validate the effectiveness of the proposed methods on data from different sensors, the Pavia University HSI dataset was added as new experimental data. The Pavia University dataset was captured by the ROSIS sensor during a remote sensing survey of the University of Pavia in Italy in 2003. The original image has a size of 610 × 340 pixels, with a ground sampling distance of 1.3 m per pixel. The wavelength range spans from 430 to 860 nm, and the original dataset consists of 115 bands. Among these, 22 bands are water vapor absorption bands, which are removed, leaving 93 bands. The spectral imager in the dataset captures 115 bands in the wavelength range of 0.43–0.86 μm continuously, but 12 bands are discarded due to noise interference. Therefore, images are usually generated using the remaining 103 spectral bands. The content of the dataset includes nine classes of ground objects: asphalt, grass, gravel, trees, metal plate, soil, asphalt, bricks, and shadows. There are a total of 42 776 labeled sample instances for the objects in the dataset. From Table VII, it can be seen that for HSI data from different data sources, the three proposed methods still achieve the highest accuracy.

E. Ablation Experiment

Table VIII shows the progressive improvement in accuracy achieved through the implementation of a proposed algorithm, along with various additional modules, for image classification tasks in different datasets.

In the Yangtze River Delta dataset, the baseline algorithm achieved an accuracy of 81.08%. The proposed one-time HRC-MS3 algorithm improved this accuracy to 84.39% without any additional modules. Subsequently, by incorporating the iterative sample expansion module, the accuracy further increased to 85.45%. Finally, after adding the WDL module, the accuracy showed a slight decrease to 85.23%, but it still remained higher than that of the baseline algorithms.

For the Dafeng Natural Reserve dataset, the baseline algorithm achieved an accuracy of 83.27%. The one-time HRC-MS3 algorithm improved this accuracy to 87.31% without any additional modules. With the inclusion of the iterative sample expansion module, the accuracy further improved to 89.16%. After adding the WDL module, the accuracy reached 89.35%.

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TABLE VIII
ABLATION ANALYSIS

Algorithm Description	Baseline	√			
	One-time HRC-MS3		√		√
	Iterative Sample Expansion			√	√
	multi-scale Weighted Dictionary Learning				√
Datasets	Yangtze River Delta (OA%)	81.08	84.39	**85.45**	85.23
	Jiangsu Dafeng Natural Reserve (OA%)	83.27	87.31	89.16	**89.35**
	Yellow River Delta (OA%)	83.17	86.49	86.83	**87.06**

In the case of the Yellow River Delta dataset, the baseline algorithm achieved an accuracy of 83.17%. The one-time HRC-MS3 algorithm improved this accuracy to 86.49% without any additional modules. With the incorporation of the iterative sample expansion module, the accuracy increased to 86.83%. Finally, after adding the WDL module, the accuracy showed a slight improvement to 87.06%. These results demonstrate the incremental enhancements in accuracy achieved by the proposed algorithm and different modules for image classification tasks.

In summary, it is indicated by the experimental results that the one-time HRC-MS3 algorithm is effective across three different datasets, and a positive impact is observed on accuracy improvement due to the iterative sample expansion module. The WDL module exhibits varying effects on different datasets but is generally noted to contribute to accuracy improvement.

VI. CONCLUSION

A novel approach is proposed that combines SP segmentation and multiclassifier EL to address the limited availability of labeled samples in coastal wetland land cover classification. Initially, the application of SP segmentation techniques is employed to partition unknown samples into multiple homogeneous regions, thereby facilitating the effective capture of spatial information pertaining to land cover. Subsequently, a multiclassifier EL strategy is employed within these regions to process the samples, effectively leading to a reduction in classification errors and an improvement in accuracy. To enhance the performance of SSL, a sample iteration selection metric is introduced to optimize the training samples based on the consistency of sample types within homogeneous regions and the results obtained from the multiclassifier ensemble, thus enhancing the reliability of pseudo-labels. Additionally, multiscale SP segmentation is utilized to augment the ensemble strategy for samples, reducing the necessity for hyperparameter adjustments and increasing the automation and reliability of the model. The superior performance of proposed HRC-MS3 and IMSE-WDL in wetland classification is demonstrated by experimental results using hyperspectral datasets from China, which can outperform state-of-the-art methods. Its extensibility is showcased by achieving better results on multiple machine learning models with new samples. Moreover, its generalizability is evidenced by the outstanding performance on the Botswana dataset. Overall, the proposed HRC-MS3 and IMSE-WDL can improve wetland classification accuracy and efficiency, suggesting opportunities for future research and practical applications in this domain. The proposed algorithm in the article has certain advantages in solving the problem of coastal wetland land cover classification, but it may be overly dependent on high-quality segmentation data and the performance of multiclassifier EL algorithms. If the segmentation quality is poor or the performance of the classifiers is limited, it will affect the overall results. Meanwhile, the sample iteration selection criterion may need to be adjusted to better adapt to different wetland environments. SSL may have limitations when utilizing unlabeled data, as this data may contain erroneous information. To address these issues, future research can incorporate advanced SSL models as base classifiers, such as GANs or self-supervised learning methods, to better exploit and utilize unlabeled sample information, improve the accuracy of pseudo-labels, and further enhance the model's generalization performance in small sample scenarios.

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REFERENCES

[1] Z. Xie, J. Hu, X. Kang, P. Duan, and S. Li, "Multilayer global spectral–spatial attention network for wetland hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518913.

[2] Z. Huang and X. Na, "Wetland mapping based on feature optimization and Google Earth Engine in Zhalong Nature Reserve," *J. Appl. Remote Sens.*, vol. 16, no. 4, p. 48501, Oct. 2022.

[3] H. Su, F. Shao, Y. Gao, H. Zhang, W. Sun, and Q. Du, "Probabilistic collaborative representation based ensemble learning for classification of wetland hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509517.

[4] T. Schmid, M. Koch, and J. Gumuzzio, "Multisensor approach to determine changes of wetland characteristics in semiarid environments (Central Spain)," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 11, pp. 2516–2525, Nov. 2005.

[5] H. Lu et al., "A probabilistic sample boosting approach with adaptive representation coefficient consistency for China coastal wetland land cover classification using GF-5 hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521918.

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[6] Y. Gao et al., "Hyperspectral and multispectral classification for coastal wetland using depthwise feature interaction network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512615.

[7] K. Liu et al., "Mapping coastal wetlands using transformer in transformer deep network on China ZY1-02D hyperspectral satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3891–3903, 2022.

[8] B. Du et al., "Mapping wetland plant communities using unmanned aerial vehicle hyperspectral imagery by comparing object/pixel-based classifications combining multiple machine-learning algorithms," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8249–8258, 2021.

[9] Q. Liu, J. Peng, G. Zhang, W. Sun, and Q. Du, "Deep contrastive learning network for small-sample hyperspectral image classification," *J. Remote Sens.*, vol. 3, Jan. 2023, Art. no. 0025.

[10] Q. Liu et al., "Refined prototypical contrastive learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506214.

[11] J. Feng et al., "Class-aligned and class-balancing generative domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509617.

[12] J. Feng, G. Bai, D. Li, X. Zhang, R. Shang, and L. Jiao, "MR-selection: A meta-reinforcement learning approach for zero-shot hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500320.

[13] P. K. Mallapragada, R. Jin, A. K. Jain, and Y. Liu, "SemiBoost: Boosting for semi-supervised learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 31, no. 11, pp. 2000–2014, Nov. 2009.

[14] J. Li, S. Wu, C. Liu, Z. Yu, and H.-S. Wong, "Semi-supervised deep coupled ensemble learning with classification landmark exploration," *IEEE Trans. Image Process.*, vol. 29, pp. 538–550, 2020.

[15] H. Wu and S. Prasad, "Semi-supervised deep learning using pseudo labels for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1259–1270, Mar. 2018.

[16] X. Hu, T. Zhou, and Y. Peng, "Semisupervised deep learning using consistency regularization and pseudolabels for hyperspectral image classification," *J. Appl. Remote Sens.*, vol. 16, no. 2, p. 26513, May 2022.

[17] Y. Ding, X. Zhao, Z. Zhang, W. Cai, N. Yang, and Y. Zhan, "Semi-supervised locality preserving dense graph neural network with ARMA filters and context-aware learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511812.

[18] J. Zhong, W. Xie, Y. Li, J. Lei, and Q. Du, "Characterization of background-anomaly separability with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6017–6028, Jul. 2021.

[19] A. Qin et al., "Distance constraint-based generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5511416.

[20] X. Wang, K. Tan, Q. Du, Y. Chen, and P. Du, "CVA2E: A conditional variational autoencoder with an adversarial training process for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5676–5692, Aug. 2020.

[21] Y. Su, J. Li, A. Plaza, A. Marinoni, P. Gamba, and S. Chakravortty, "DAEN: Deep autoencoder networks for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4309–4321, Jul. 2019.

[22] Y. Zhan, D. Hu, Y. Wang, and X. Yu, "Semisupervised hyperspectral image classification based on generative adversarial networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 212–216, Feb. 2018.

[23] J. Feng, H. Yu, L. Wang, X. Cao, X. Zhang, and L. Jiao, "Classification of hyperspectral images based on multiclass spatial–spectral generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5329–5343, Aug. 2019.

[24] T. Dam, S. G. Anavatti, and H. A. Abbass, "Mixture of spectral generative adversarial networks for imbalanced hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[25] W. Yu, M. Zhang, Z. He, and Y. Shen, "Convolutional two-stream generative adversarial network-based hyperspectral feature extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5506010.

[26] A. Sha, B. Wang, X. Wu, and L. Zhang, "Semisupervised classification for hyperspectral images using graph attention networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 157–161, Jan. 2021.

[27] X. He, Y. Chen, and P. Ghamisi, "Dual graph convolutional network for hyperspectral image classification with limited training samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502418.

[28] Y. Ding et al., "Self-supervised locality preserving low-pass graph convolutional embedding for large-scale hyperspectral image clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536016.

[29] F. Li, D. A. Clausi, L. Xu, and A. Wong, "ST-IRGS: A region-based self-training algorithm applied to hyperspectral image classification and segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 3–16, Jan. 2018.

[30] L. Zhao, W. Luo, Q. Liao, S. Chen, and J. Wu, "Hyperspectral image classification with contrastive self-supervised learning under limited labeled samples," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[31] J. Zhu, K. Tan, and Q. Du, "Tri_training for remote sensing classification based on multi-scale homogeneity," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 3055–3058.

[32] J. Li, H. Zhang, and L. Zhang, "Efficient superpixel-level multitask joint sparse representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 10, pp. 5338–5351, Oct. 2015.

[33] S. Jia, X. Deng, J. Zhu, M. Xu, J. Zhou, and X. Jia, "Collaborative representation-based multiscale superpixel fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7770–7784, Oct. 2019.

[34] H. Xu, H. Zhang, W. He, and L. Zhang, "Superpixel-based spatial–spectral dimension reduction for hyperspectral imagery classification," *Neurocomputing*, vol. 360, pp. 138–150, Sep. 2019.

[35] W. Li, Q. Du, and M. Xiong, "Kernel collaborative representation with Tikhonov regularization for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 1, pp. 48–52, Jan. 2015.

[36] L. Jiao, W. Sun, G. Yang, G. Ren, and Y. Liu, "A hierarchical classification framework of satellite multispectral/hyperspectral images for mapping coastal wetlands," *Remote Sens.*, vol. 11, no. 19, p. 2238, Sep. 2019.

[37] X. Sun, L. Guo, W. Zhang, Z. Wang, and Q. Yu, "Small aerial target detection for airborne infrared detection systems using LightGBM and trajectory constraints," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9959–9973, 2021.

[38] Y. Cui, C. Shao, L. Luo, L. Wang, S. Gao, and L. Chen, "Center weighted convolution and GraphSAGE cooperative network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508216.

[39] K. C. A. Kumar and C. D. Vleeschouwer, "Discriminative label propagation for multi-object tracking with sporadic appearance features," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 2000–2007.

[40] Z.-H. Zhou and M. Li, "Tri-training: Exploiting unlabeled data using three classifiers," *IEEE Trans. Knowl. Data Eng.*, vol. 17, no. 11, pp. 1529–1541, Nov. 2005.

[41] A. Samat, E. Li, P. Du, S. Liu, and J. Xia, "GPU-accelerated CatBoost-forest for hyperspectral image classification via parallelized mRMR ensemble subspace feature selection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3200–3214, 2021.

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